

Overview of Maritime Datasets for Vision-Based Navigation and Detection

The table below summarizes a comprehensive list of maritime-related datasets used in scientific research for tasks such as object detection, autonomous navigation, and maritime surveillance. Datasets span various sensor modalities (camera, infrared, LiDAR, radar, AIS, satellite imagery, etc.) and cover diverse maritime environments (open sea, coastal, port, rivers). Each dataset is accompanied by a reference to a scientific paper or article introducing or using it, along with links to the dataset and publication for further details.

Dataset (Year)	Sensors	Description & Applications	Dataset Link	Paper Reference
Singapore Maritime Dataset (SMD) (2017)	Visible camera (RGB) and Near-IR camera	81 video sequences (1080p) captured on-shore and on-board around Singapore waters, with annotations for object detection and tracking of vessels ¹ ² . Collected under various illumination (day, night, haze) for maritime surveillance and collision avoidance research ³ ⁴ .	Website	Prasad <i>et al.</i> , IEEE T-ITS 2017 ⁵ ⁶
MARVEL (2016)	Visible images (web-crawled photos)	Large-scale image dataset of ships for vessel identification and fine-grained classification . Contains ~2 million user-uploaded ship images categorized into 109 vessel types (grouped into 26 super-classes) ⁷ . One ship per image with high quality, suitable for training recognition models ⁸ ⁹ .	GitHub	Gundogdu <i>et al.</i> , ACCV 2016 ¹⁰ ¹¹
MarDCT (ARGOS-Venice) (2015)	Visible surveillance cameras (video)	Early public dataset for maritime surveillance, supporting boat classification and tracking . Includes 6,742 images and 16 videos of Venice waterways, with annotated boats (image resolution ~800×240) ¹² . Developed from a 24/7 harbor monitoring system in Venice ¹³ . Useful for vision-based USV navigation research.	<i>By request</i> (ARGOS system)	Bloisi <i>et al.</i> , IEEE AVSS 2015 ¹⁴ ¹²

Dataset (Year)	Sensors	Description & Applications	Dataset Link	Paper Reference
SeaDronesSee (2022)	UAV-mounted cameras (RGB + multi-spectral)	High-altitude drone dataset for search & rescue: >54,000 video frames with ~400k annotations of swimmers (with/without life vests), boats, etc., for object detection and tracking ¹⁵ . Includes meta-data (altitude, camera angle) and multi-spectral images (RedEdge/ NIR channels) to aid robust human detection in open water ¹⁶ ¹⁷ .	Project Site	Varga <i>et al.</i> , WACV 2022 ¹⁵ ¹⁷
KOLOMVERSE (2022)	Multi-camera video (4× 4K RGB cameras)	A large-scale maritime object dataset from Korean waters – 186,419 4K images extracted from 5,845 hours of video ¹⁸ . Collected via a 4-channel CCTV system on 17 ships across 21 sea zones ¹⁹ . Provides bounding boxes for ships and obstacles, aimed at improving maritime collision avoidance and monitoring systems ²⁰ ²¹ .	GitHub	Jang <i>et al.</i> , arXiv/ NeurIPS Datasets 2023 ²² ²³
M2SODAI (2023)	Aerial RGB camera + Hyperspectral sensor	Multi-modal aerial dataset for ship and debris detection : synchronized high-altitude RGB and hyperspectral images with bounding-box annotations ²⁴ . Contains ~5,764 instances of ships and floating objects, used to demonstrate that fusing hyperspectral data can reduce false detections from sun glint and waves ²⁵ ²⁴ .	Project Page	Jang <i>et al.</i> , NeurIPS D&B 2023 ²⁶ ²⁴

Dataset (Year)	Sensors	Description & Applications	Dataset Link	Paper Reference
PoLaRIS (Pohang Canal) (2024)	Multi-modal: RGB camera, Thermal IR, LiDAR, Radar	New multi-sensor dataset for obstacle detection and tracking in a narrow waterway (Pohang canal). Provides synchronized RGB and thermal imagery with corresponding 3D LiDAR point clouds and marine radar scans, all annotated for hazards (objects as small as 10×10 pixels) ²⁷ ²⁸ . Enables evaluation of multi-modal sensor fusion for USV autonomous navigation under low-visibility and long-range conditions.	Dataset Site	Choi <i>et al.</i> , arXiv 2024 ²⁷ ²⁸
Pohang Canal Dataset (2023)	Multi-modal: 3× LiDAR, Radar, stereo RGB, IR, 360° camera, GPS/IMU	A rich multimodal dataset from KAIST for autonomous ship navigation in restricted waters ²⁹ ³⁰ . Collected along a 7.5 km canal and harbor route (day & night) with three LiDARs (64ch + 2×32ch), a marine radar, stereo cameras, a thermal camera, and a 6-view omnidirectional camera ³⁰ . Includes sensor calibrations and a SLAM-based trajectory; used to develop and benchmark USV navigation and obstacle avoidance algorithms ³¹ ³² .	Site/AWS	Chung <i>et al.</i> , IJRR 2023 ³³ ³²
Seagull UAV Dataset (2017)	Multiple cameras: Visible, Near-IR, Thermal IR, Hyperspectral	Multi-camera, multi-spectrum airborne surveillance dataset ³⁴ . Video sequences captured from a fixed-wing UAV over the sea, containing various targets (boats, life rafts, oil-spill simulating objects) ³⁴ . Provides aligned imagery in visible, NIR, and LWIR infrared, plus a hyperspectral sensor (visible range) for research on maritime search & rescue and environmental monitoring ³⁵ .	<i>Available on request</i> (VisLab)	Ribeiro <i>et al.</i> , IEEE T- CSVT 2017 ³⁶ ³⁴

Dataset (Year)	Sensors	Description & Applications	Dataset Link	Paper Reference
Piraeus AIS Dataset (2017–19)	AIS (Automatic Identification System) ship position feeds	A large-scale AIS data collection for vessel traffic analytics. Logs over 244 million AIS messages (≈ 2.5 years) from ships around the Port of Piraeus, Greece ³⁷ . Includes time-stamped positions and correlated weather and area data, enabling studies in maritime traffic patterns, anomaly detection, and route planning ³⁷ ³⁸ . Publicly released for big-data maritime research.	Zenodo	Tritsarolis <i>et al.</i> , <i>Data in Brief</i> 2022 ³⁷ ³⁹
ABOships (2021)	Visible images (frames extracted from video)	A curated image dataset for maritime vessel detection in both inshore and offshore scenes. Contains 9,880 images of maritime scenes with objects labeled in 11 categories: 9 types of vessels (e.g. cargo ship, ferry, motorboat, etc.), seamarks, and miscellaneous floaters ⁴⁰ . Images cover diverse backgrounds, weather, lighting and are densely annotated (avg >4 objects/image) ⁴¹ ⁴⁰ . Provides a benchmark for deep-learning detection models in marine environments.	Zenodo	Iancu <i>et al.</i> , <i>Remote Sensing</i> 2021 ⁴⁰ ⁴¹
InaTechShips (2025)	Visible images (photos, various sources)	Massive dataset of ship images for detection and classification, featuring 3+ million high-quality images of maritime vessels in diverse scenes ⁴² . Images were automatically labeled into numerous vessel categories, facilitating training of robust models for maritime surveillance, traffic monitoring, and security applications ⁴² ⁴³ . Accompanied by baseline models and an open-source toolkit.	GitHub	Teixeira <i>et al.</i> , <i>Ocean Eng.</i> 2025 (accepted) ⁴² ⁴⁴

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SeaShips (MCVWT) (2022)	Visible images (camera snapshots)	Medium-scale image dataset for ship detection and classification, used in vision benchmarks. Contains ~6,979 images (1920×1080) of vessels spanning 6 classes (fishing boat, bulk cargo, container ship, etc.) ⁴⁵ ⁴⁶ . All ships are labeled with bounding boxes; conditions are mostly clear or hazy weather on calm seas ⁴⁵ ⁴⁷ . Suitable for training models focused on maritime traffic surveillance and vessel type recognition.	Roboflow	Soloviev <i>et al.</i> , ICMEW 2020 ⁴⁸ ⁴⁹
MassMIND (2023)	Long-Wave Infrared (Thermal camera)	An infrared maritime dataset for obstacle detection in all weather, geared towards autonomous navigation. Provides 2,916 thermal images of marine scenes with diverse conditions (night/day, fog, rough seas, etc.) ⁵⁰ . Annotations include semantic masks and instance segmentations for 7 classes (sky, water, bridge, vessel/ obstacle, etc.), enabling training of deep models to detect objects and horizon in thermal imagery ⁵¹ ⁵² .	GitHub	Nirgudkar <i>et al.</i> , IJRR 2023 ⁵³ ⁵²
MUSSID (2022)	Visible images (HD RGB photos)	The “Maritime Unmanned Surface vehicle Sea Image Dataset” for sea horizon line detection and navigation ⁵⁴ . Consists of 2,673 high-resolution images from various geographies and seasons, capturing a wide range of sea states, lighting, and weather conditions ⁵⁵ . Each image is annotated with the horizon line (and other linear features) to train and evaluate algorithms for robust horizon detection in maritime scenes (critical for vision-based navigation) ⁵⁶ ⁵⁵ .	Kaggle	Hashmani <i>et al.</i> , JMSE 2022 ⁵⁶ ⁵⁷

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MariShipSeg HEU (2020)	Visible images	A hybrid dataset for ship segmentation and classification , created at Harbin Engineering Univ. It contains 3,560 images, pooled from prior sets (incl. ~10% from SMD and 25% from MarDCT) and internet photos ⁵⁸ ⁵⁹ . All images are manually labeled for ship presence and pixel-wise ship masks. Ships are generally large and centered (high-quality, curated images) ⁶⁰ ⁶¹ . Useful as a starter dataset for maritime AI (less diverse but with clean annotations).	GitHub	Zhang <i>et al.</i> , Image Vis. Comput. 2020 ⁶² ⁶³
VAIS (2015)	Visible + Infrared camera images	A bi-spectral image dataset for small vessel detection in surveillance. Contains 2,014 training images of maritime scenes captured in both visible and IR spectra ⁶⁴ ⁶⁵ . Each image has at most one boat (annotated by bounding box), enabling evaluation of cross-spectrum detection algorithms ⁶⁵ . Though limited in diversity (single “boat” class and often one vessel per image), it demonstrated the benefit of IR for detecting ships under challenging lighting ⁶⁶ ⁶⁷ .	Project	Zhang <i>et al.</i> , CVPRW 2015 ⁶⁸

Dataset (Year)	Sensors	Description & Applications	Dataset Link	Paper Reference
HRSC2016 (2016)	Satellite imagery (high-res optical)	High-Resolution Ship Collections 2016 – a widely used satellite image dataset for ship detection. Comprises imagery from 6 famous ports with ships annotated by oriented bounding boxes in 22 distinct ship classes ⁶⁹ . Enables testing of algorithms on multi-scale, multi-angle ship instances in complex backgrounds (harbors). Often used as a benchmark for rotation-aware object detectors in remote sensing ⁶⁹ ⁷⁰ .	Kaggle	Liu <i>et al.</i> , ICIIP 2017 (HRSC2016 paper) ⁷¹ ⁶⁹

Each of these datasets is publicly available (fully or upon request) and backed by a publication, providing a foundation and benchmarks for developing and evaluating maritime computer vision and autonomous navigation algorithms. The variety of sensors and scenarios covered allow researchers to address tasks from ship detection in port surveillance to obstacle avoidance for autonomous surface vehicles, across both electro-optical and radar/IR modalities. All sources and links are provided for access to the data and the respective papers.

¹ ³ ⁵ ¹⁴ ³⁶ ^{dilipprasad} - Singapore Maritime Dataset

<https://sites.google.com/site/dilipprasad/home/singapore-maritime-dataset>

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